

Threat Intelligence

Estimated time needed: 5 minutes

Introduction

Threat intelligence is collating and analyzing information about potential attacks that could threaten an organization. This includes data about cybercriminals' methods, tactics and strategies, identified patterns, and indicators of compromise. It aids in preemptively defending against cybersecurity threats, strengthening the organization's security posture.

Threat Intelligence Types

Threat intelligence can be categorized into three primary types, each offering a unique perspective and valuable insights to safeguard an organization's cyber domain.

Source	Description
Behavioral Threat Intelligence	This type of intelligence focuses on understanding attackers' patterns, methodologies, and tactics. By analyzing their behaviors, security teams can anticipate potential attacks and develop strategies to counter them. It's a proactive approach to detect anomalies or deviations in network behavior that could indicate a cybersecurity threat.
Reputational Threat Intelligence	Reputational intelligence involves tracking the reputation of various IP addresses, domains, and URLs. It considers the past behaviors and actions associated with these entities. It assesses the risk they pose based on their history. This intelligence is crucial in avoiding interaction with malicious domains or IP addresses.
Threat Data	This is raw, unprocessed information about potential threats. It can be derived from various sources, such as logs, alerts, and packet captures. Though it's not immediately actionable, this data is essential for conducting deeper analyses and generating helpful threat intelligence. It forms the base upon which other forms of threat intelligence are built.

Sources for Threat Intelligence

Threat intelligence comes from several key sources. These sources enable organizations to stay aware of the ever-evolving landscape of cyber threats.

Source	Description	Intelligence Examples
Open-Source Threat Feeds	These are publicly accessible sources of threat data, available to anyone with an internet connection. They include blogs, forums, and social media platforms where threat actors may share or discuss their methods. While these sources can provide valuable information, they require diligent monitoring and analysis to separate the valid data from the noise.	Examples of open-source threat feeds include the likes of ThreatConnect, AlienVault Open Threat Exchange (OTX), Twitter feeds like @malwrhunterteam, and cybersecurity blogs such as Krebs on Security and Dark Reading.
Proprietary Threat Feeds	These are owned and operated by private organizations, usually cybersecurity companies. They typically offer more comprehensive and detailed information about threats, often derived from their networks and clientele. Access to these feeds usually requires a subscription, but they can provide valuable, actionable insights unavailable through open sources.	Examples of proprietary threat feeds include services offered by cybersecurity companies like Symantec's DeepSight Intelligence, IBM's X-Force Exchange, and CrowdStrike's Falcon X Threat Intelligence.
Deep and Dark Web	These areas of the internet are not indexed by traditional search engines and are often associated with illegal activities. Threat actors often communicate, plan, and trade tools on deep and dark web platforms. Intelligence gathered from these sources can provide early warning of planned attacks or reveal new attack methods and tools. However, gathering intelligence from these sources requires specialized tools and expertise.	Specific examples of the deep and dark web include platforms like the infamous Silk Road (now defunct), AlphaBay, and hidden forums on the Tor network where cybercriminals can communicate, trade illicit goods, plan attacks, and share malware or exploit kits.
Industry Reports and Threat Bulletins	Cybersecurity entities and research organizations publish reports and bulletins detailing current threat trends, new vulnerabilities, and recent cyberattacks. These documents can provide valuable insights into the global cybersecurity landscape.	Examples of industry reports and threat bulletins include IBM's X-Force Threat Intelligence Index, Verizon's Data Breach Investigations Report, and bulletins from the United States Computer Emergency Readiness Team (US-CERT).
Government and Regulatory Bodies	National cybersecurity agencies and international regulatory bodies often release threat advisories and alerts. These can be particularly useful in identifying state-sponsored threats and large-scale cybercrime activities. National cybersecurity agencies and international regulatory bodies often release threat advisories and alerts. These can be particularly useful in identifying state-sponsored threats and large-scale cybercrime activities.	Examples of advisories and alerts from government and regulatory bodies include the Cybersecurity and Infrastructure Security Agency (CISA) alerts, Federal Bureau of Investigation (FBI) cybercrime reports, warnings from the National Cyber Security Centre (NCSC) in the UK, and bulletins from the European Union Agency for Cybersecurity (ENISA).
Incident Response (IR) Teams	The internal or external IR teams responding to cyber incidents within the organization can provide first-hand threat intelligence. Their interactions with the threat actors and understanding of the methods employed in an attack can offer unique insights.	Examples of intelligence gathered from Incident Response (IR) teams may include details of a ransomware attack's methodology, the IP addresses used in a phishing campaign, or the unique signatures of a newly encountered malware.

Honeypots and Deception Technology	These are decoy systems or networks designed to attract and engage attackers. The attackers' activities on these systems are monitored and analyzed to gain intelligence about their tactics, techniques, and procedures (TTPs).	Examples of threat intelligence gathered from honeypots and deception technology could include discovering novel malware variants, delineating attacker behavior patterns, and identifying previously unknown attack vectors or techniques.
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